Md Shohanoor Rahman

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Research Interests

Trustworthy and explainable machine learning, especially testing whether explanations are faithful; concept-based models; diffusion models and their evaluation; medical image analysis, especially dermoscopy.

Technical Summary

- Research experience in counterfactual concept audits of melanoma classifiers using diffusion models.
- Research experience in concept-conditioned diffusion models (DDPM, DDIM, CFG) for dermoscopy images.
- Research experience in testing TCAV and attribution methods against controls with a known answer.
- Research experience in explainable ML for clinical data with SHAP and LIME (two IEEE papers).
- Research experience in GAN-based synthetic tabular data generation and its statistical evaluation.
- Hands-on experience in PyTorch training of CNN and ViT models on multi-GPU (2× T4) and TPU v5e-8.
- Hands-on experience in reproducible research: rules fixed in code before each run, public code.
- Hands-on experience in industry software engineering (3.5+ years): Spring Boot, Kafka, Kubernetes, Azure.

Education

Rajshahi University of Engineering & Technology (RUET) — Rajshahi, Bangladesh
B.Sc. in Computer Science & Engineering — *Jan 2017 – Aug 2023*

- **Thesis:** *Generative Adversarial Network based Synthetic Data Generation System.* Supervisor: Barshon Sen. Github: UG-Thesis
- **Relevant coursework**

Artificial Intelligence	Applied Statistics and Queuing Theory	Operating Systems
Neural Networks and Fuzzy Systems	Discrete Mathematics	Parallel and Distributed Processing
Data Mining	Numerical Methods	Computer Architecture and Design
Digital Image Processing	Data Structure	Software Engineering
Digital Signal Processing	Computer Algorithms	Information System Analysis and Design
Vector Analysis and Linear Algebra	Database Systems	Technical Writing and Presentation

Publications

Peer-reviewed conference papers

[C1] **Md. Shohanoor Rahman**, Abid Hasan Mahmud, S. M. Mehedy Hasan. "Interpretable Multi-Class Obesity Classification Using Bayesian Optimization and SHAP Insights." In *Proc. 28th International Conference on Computer and Information Technology (ICCIT)*, IEEE, Cox's Bazar, Bangladesh, Dec 2025. DOI: 10.1109/ICCIT68739.2025.11491728. *Presented at the conference.*

[C2] **Md. Shohanoor Rahman**, Abid Hasan Mahmud, S. M. Mehedy Hasan. "Explainable Machine Learning for Chronic Kidney Disease Prediction with SHAP and LIME." In *Proc. 28th International Conference on Computer and Information Technology (ICCIT)*, IEEE, Cox's Bazar, Bangladesh, Dec 2025. DOI: 10.1109/ICCIT68739.2025.11490146.

Journal manuscript

[J1] **Md. Shohanoor Rahman**, Md. Hasibul Islam, Md Kishor Morol, Moinul Hossain, Dip Nandi, Liew Tze Hui. "Validating concept-level audits of melanoma classifiers with diffusion counterfactuals and matched positive controls." [**Status: Under review, *Scientific Reports*, submitted Oct 2026**]. Code and result tables: github.com/srahman7900/melanoma-concept-audit.

Research Experience

Validating concept-level audits of melanoma classifiers [J1] — Remote
First and corresponding author. Supervisors: Md Kishor Morol and Liew Tze Hui — *Dec 2025 – Present*

- Conceived the study, designed the method, wrote all code, ran all experiments on free Kaggle compute (2× T4, TPU v5e-8), did the analysis and wrote the first draft.
- Built a counterfactual concept audit from a concept-conditioned DDPM (128 px, ISIC 2020) in which each of the seven checklist concepts is a knob from 0 to 1. Found a signal-leak bias in the start noise ($R^2 = 0.90$) and removed it with a colour-calibrated start noise (colour outliers fell from 78.5–84.8% to 6.6–13.3%); an independent evaluator confirmed 4 of 7 knobs as specific.
- Tested the audit on 38 classifiers: six ISIC 2020 melanoma classifiers (ResNet-50, EfficientNet-B0, ViT-B/16), one Derm7pt classifier and 31 controls with a known answer. Fixed each decision rule in code before its run and reported the rule that failed; matched controls then passed 16 of 16, and independently labelled controls passed 12 of 12 for three concepts under a SHA-256-fingerprinted rule.

- Two TCAV implementations failed a permuted-label negative control (0.834 and 0.889; limit 0.25). Observational attribution ranked vascular structures first for all six melanoma classifiers; the validated audit found a large pigmentation effect (+2.77 logits) and small vascular (+0.29) and pigment-network (+0.05) effects.

Explainable machine learning for clinical data Bangladesh
Group lead. Advisor: S. M. Mahedy Hasan, Assistant Professor, CSE, RUET *Jan 2025 – Dec 2025*

- Led a research group while working full time: literature review, all code, and the manuscripts of two first-author IEEE ICCIT 2025 papers [C1, C2].
- Built SHAP and LIME pipelines for chronic kidney disease prediction (4,921 patient records) and benchmarked eleven classifiers for six-level obesity classification on a new dataset of 650 participants collected in Bangladesh.

Undergraduate thesis: synthetic tabular data with GANs RUET, Rajshahi
Supervisor: Barshon Sen, Assistant Professor, CSE, RUET *Completed Nov 2022*

- Generated synthetic data with CTGAN for the Taiwanese Bankruptcy Prediction dataset (6,819 firms, 3.23% bankrupt) in a two-person project; responsible for the literature review, exploratory analysis and results analysis (statistical tests, PCA, mutual information, train-on-synthetic, test-on-real evaluation).
- Presented each phase of the work to department faculty, students and external guests in the thesis seminar.

Professional Experience

Brain Station 23 PLC Dhaka, Bangladesh
Software Engineer *Mar 2023 – Present*

- **IRIS platform, Tennant Company (USA).** Azure telemetry for autonomous cleaning machines; contributed to the Overview dashboard, Tech Hub (semantic search, Azure AI Search) and customer onboarding.
- **AB Direct (AB Bank) and NCC Always (NCC Bank).** Features for two live internet banking platforms: transfers, bill payment, deposits, service requests. *Java, Spring Boot, Kafka, PostgreSQL, Redis, Kubernetes.*
- **Pocket (ABG Technologies, Bashundhara Group).** Admin, customer and merchant portals of a mobile wallet licensed by Bangladesh Bank. *Spring Boot, Angular, Kafka.*

Technical Skills

- **Machine learning:** PyTorch, TensorFlow, scikit-learn, XGBoost, CatBoost; diffusion models (DDPM, DDIM, classifier-free guidance), GANs, concept bottleneck models, CNN and ViT training.
- **Explainability and evaluation:** TCAV and Captum, SHAP, LIME; interventional and matched-control evaluation; non-parametric statistics (Wilcoxon, Benjamini-Hochberg, bootstrap).
- **Programming and tools:** Python, Java, TypeScript, C/C++, SQL; Git, LaTeX, JIRA, Confluence.
- **Systems:** Spring Boot, Angular, Kafka, PostgreSQL, Redis, Kubernetes, Microsoft Azure (Service Bus, AI Search).
- **Languages:** Bengali (native), English (professional working proficiency).

Additional Training, Projects and Activities

- **DeepLearning.AI (Coursera):** Generative Adversarial Networks Specialization; four Deep Learning Specialization courses (neural networks, optimization, CNNs, sequence models).
- **Course projects:** conditional GAN (pix2pix) for aerial-image-to-map translation; CycleGAN for unpaired image translation; U-Net for biomedical image segmentation.
- **RUET Analytical Programming Lab:** member for two years; took part in programming contests and helped organise a coding contest.

References

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